**Solidarity for Agri-Intelligence and Circular Economy (SAICE) – Solidarité pour Agri-Intelligent et Economy Circulaire (SAIEC)**

We provide our meaningful to contribute to economic resilience, climate change and sustainable Impacts in Developing Countries

# PAGES - Track 1

### ''Precision Agriculture for Green Economy and Sustainability''

## Rationale

Precision Agriculture for Green Economy and Sustainability (PAGES) is a programme owned by IFSS and partners.
PAGES main goal is to support farmers and Agri-SMEs in developing countries. In great Lake Region (GLR) of Africa, PAGEs is empowering communities, including smallholders' farmers and Agri-SMEs with small Capital in great lake region of Africa.

The programme campaigns communities from the farm-to- Folk, by providing technical assistance at each stage of food production, storage, market and financial linkages.
The assistance includes technologies deployment, training, coaching and develop with its beneficiaries a strategic partnership that leads to sustainable growth, financial and market inclusions.

With its beneficiaries, PAGES – monitors and evaluate the effectiveness, cost efficiency and the improvements towards sustainable impacts.

**PAGES** is committed to support his partners, including public and private organizations, small and Medium Businesses, and vulnerable communities.
PAGES technologies are friendly- environment and easy use and can adapt languages – No need to have a lot of studies, a little guide is enough to become autonomous.

## Tools and Assistance (T&A)

### Smart Market Matchmaker

The tools are specialized to link producers to the market where farmers are linked to potential buyers, and they then increase the farmers power to take the decision on the price of the produce. The tools take provide information on the localization of the farmers.

**Figures/image (if possible)**

### IoT-Precision Insight Engine

This smart tool provides information on the quality of soil moisture **b**y leveraging ArcGIS, GPS data, and NDVI. With this technology, ***PAGES*** supports farmers to monitors crops diseases and the production global health and farming system.
Due to PAGES support farmers are timely informed on the general health of her farming system and increases the decision planning for his production.
This programme empowers vulnerable communities, small household farmers and increase their bargaining power on financial aspects, including insurance, financial access and the price negotiation with buyers.

***Figures – Video and Demon (if possible)***

### ERP Finance & Inventory Tracker

PAGES supports small businesses to increase the profit and to grow their businesses. PAGES automates business accounting, sales, inventory, and production with one comprehensive solution.

**Figure/video**

## Strategic Partners

Via PAGES and its technologies, IFSS is contributing to increase the food production and enable vulnerable communities, small businesses to have access to the market and financial systems. The programme enables small hold farmers and agri-businesses, Youth and Women to increase profits and be more climate – resilient.

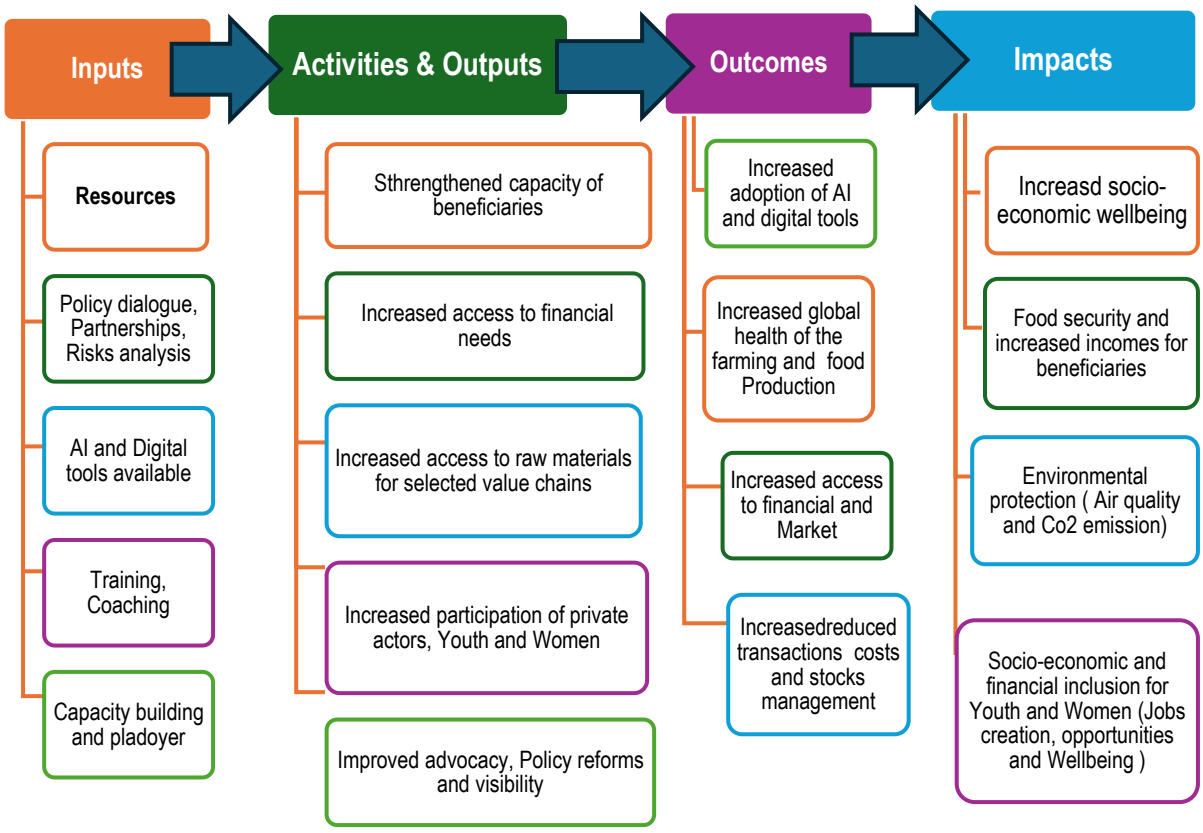
In Great Lake Region, the programme is being implemented in the partnership with Regional Research Centre for Integrated Development (RCID) Southern Kenya University (SEKU), McGill University.

**For more information about our services, partnership you contact us at -------**

**Our Communication and digital specialist will come to you as soon as possible**

**Programme Manager**

# Theory of Change for PAGES



# ICES – Track 2

**Initiative for Circular Economy and Sustainability**

## Rationale

Initiative for Circular Economy and Sustainability consist of supporting communities, municipalities and Youth organizations in charges of waste collection and management.
The goal is to enhance the efficiency of waste collection management, enabling the storage and conversion of waste into renewable energy and contributing to the Greenhouse gas emissions (Methane and Co2) reduction.
By leveraging digital tools, ICE- Plus seeks to optimize the collection routes and improve data accuracy in waste quantification. This initiative is not only to support environmental sustainability but also create employment opportunities for youth and Women.

## ICE-Plus Tools and Technical Assistance

ICE+ tools include IoT sensors, AI, and GPS to optimize collection by monitoring bin levels in real-time, reducing fuel consumption and operational costs.

These systems allow for dynamic routing and automated sorting, improving recycling rates and reducing contamination. Advantages of these systems include improved efficiency, lower emissions, and data-driven decision-making.

In addition, the tools are environment-friends and easy for use. Its use does not require a lot of knowledge in sciences or technologies.

**IoT Sensors (Smart Bins):**

This tool is relevant as it enables to reduce unnecessary, costly, and high-emission trips while collecting waste. The tool is equipped with sensors monitor fill levels, signaling trucks only when bins are full.

Picture or video ( If possible)

**AI & Robotics in Sorting:**

This ICE- Plus tool is useful for waste management process with efficacy and efficiency. A Computer vision and AI-powered robots identify, classify, and separate recyclables with higher accuracy than manual labor.
Images or video (If possible)

**RFID & Asset Management:**

This tool is about tagging waste containers (bins, dumpsters) with rugged RFID tags to automate identification, monitoring, and collection.

Using RFID readers on trucks, operators can track, weigh, and log in real-time, reducing theft, optimizing routes, and supporting "pay-as-you-throw" billing.

**Data Analytics Platform:**

This tool will enable to analyze historical data to forecast future waste generation and optimize resource allocation for the coming project.

**Mobile & Digital Applications:**

The tool will enable field workers, communities and Youth organization to use applications for real-time monitoring and to receive, log, and report data from the field.

**Calculator and Machine learning**

The integration of calculators and machine learning (ML) will allow a precise estimation of carbon emissions before a project of Waste management in the municipality—including AI training itself—and enables sophisticated analysis of carbon's impact on financial markets.

The tool will also help to monitor the impact of the project, day after day, on energy generation Co2 emissions and financial impacts for beneficiaries and the best strategies for sustainability.

# Strategic partners

Southern Kenya University (SEKU), RCID, McGill University, Terre des Jeunes Togo and Ripen

# Theory of Change

